

Rohan R

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EDUCATION

Global Academy of Technology

Bachelor of Engineering in Artificial Intelligence & Data Science | GPA: 9.48/10

Bangalore, India

Expected 2026

EXPERIENCE

Software Engineer Intern

ABB India

Feb 2026 – Current

Bangalore, India

Defect Detection using AI (Industrial Quality Control)

- Built an **SSIM**-based defect detection engine for 3P/4P MCCBs; resolved 3D parallax and geometric misalignment via **ORB Feature Matching** + Homography prior to structural comparison.
- Applied **CLAHE** histogram matching for lighting-invariant SSIM and replaced **Otsu's global thresholding** with localized gradient-map analysis to eliminate false positives on defect-free units.
- Validated pipeline on the factory floor across **3P and 4P MCCB series** after iterating through 5+ approaches including template matching, skeleton comparison, and background subtraction.

OCR-based Invoice Extraction

- Built a **PaddleOCR** + **OpenCV** pipeline with dynamic ROI extraction (logo + barcode anchoring) achieving **~92% avg confidence** and **~2.5s** processing time per invoice.
- Delivered dual-format JSON (production + debug with bounding coordinates) with null-value filtering; validated across **18 stations** spanning XT1, Formula A1/A2, and XT3 product lines.

PROJECTS

Face Anti-Spoofing System | *Python, TensorFlow, Keras, OpenCV, FastAPI, Streamlit* | [GitHub](#)

- Built dual-model pipeline: fine-tuned **MobileNetV2 CNN (98%)** and **LBP+SVM (97%)** trained on CelebA-Spoof (12k+ images) covering print, replay, and 3D mask attacks.
- Decoupled **FastAPI** backend from Streamlit UI, reducing cold-start from **~18s to <2s**; built 10-frame temporal voting with scene-change detection to eliminate prediction flicker.
- Integrated **Grad-CAM++** heatmap overlays to visualize model decision-making across facial regions, enabling explainable Real vs. Spoof verdicts.

ZipRoute – AI-Powered Route Optimization | *Python, XGBoost, PaddleOCR, FastAPI* | [GitHub](#) | [Live Demo](#)

- Led ML backend in a team project; owned **XGBoost** ETA prediction pipeline trained on 144,867 trips with 24 spatio-temporal features alongside team-built **2-Opt** route optimization and **PaddleOCR** address extraction.
- Achieved **±8 min ETA accuracy** (47% over baseline) and **13.9% route distance reduction**; resolved **98.5%** of unstructured label addresses via hybrid OCR-to-geocoding pipeline.
- Engineered spatio-temporal features (stop density, time-of-day, peak-hour traffic multipliers) and hybrid two-stage geocoding (OpenRouteService → Nominatim fallback) achieving **0.98s** avg API response time.

BrainBolt – Multimodal RAG System | *Python, LangChain, FAISS, FastAPI* | [GitHub](#) | [Live Demo](#)

- Built multimodal RAG pipeline over PDFs, images, YouTube transcripts, and web links using dual-index **FAISS**; images captioned via **Gemini Vision** and embedded separately to preserve visual context.
- Powered by **Google Gemini 1.5 Pro** with 6-mode summarization and difficulty-graded quiz generation; achieved **TTFT < 5s** for text/PDF inputs via async **FastAPI** backend.
- Resolved API rate-limiting via exponential backoff and chunk-size optimization; deployed on **Hugging Face Spaces**.

TECHNICAL SKILLS

Languages: Python, C/C++, Dart, SQL

AI & Machine Learning: TensorFlow, Keras, Scikit-learn, XGBoost, OpenCV, PaddleOCR, Pandas, NumPy

Generative AI & NLP: LLMs, LangChain, FAISS, Sentence-Transformers, Pinecone, Prompt Engineering

Tools & Technologies: FastAPI, Docker, MLFlow, Git, Linux, Streamlit, Jupyter, Tableau, Power BI